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EXP 12 : END-END COMMUNICATION AT TRANSPORT LAYER

AIM:

To implement echo client server using TCP/UDP sockets.

TCP:

CODE:

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET, socket.
                                        SOCK_STREAMS)
    sever.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")
    conn, addr = server.accept()
    print(f"[SERVER] connected by {addr}")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print(f"[SERVER] Received : {data}")
            conn.send(f"server received : {data}".encod
    conn.close()
    server.close()

def start_client():
    client = socket.socket(socket.AF_INET, socket.sock_
                                        STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")
    while True:
        msg = input("Enter message (or 'quit' to 'exit'):")
```

```
    client.send(msg.encode())
    if msg.lower() == 'quit':
        break
    response = client.recv(1024).decode()
    print(f"[SERVER RESPONSE] {response}")
client.close()
if __name__ == "__main__":
    server_thread = threading.Thread(
                target = start_server,
                daemon = true
            )
    server_thread.start()
    start_client()
```

OUTPUT:

```
python practical12_tcp.py
[SERVER] Listening on 127.0.0.1.5000
[CLIENT] connected to server
[SERVER] connected by ('127.0.0.1', 50501)
Enter message (or 'quit' to exit): Computer Netw
[SERVER] Received: Computer Networks
[SERVER RESPONSE] Server received: Computer Network
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client
```

UDP:

Code:

```
import socket
import threading
def udp-server():
    s = socket.socket(
        socket.AF_INET,
        socket.SOCK_DGRAM
    )
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] client {addr}
                ended the connection.")
            break
        print(f"[SERVER] Received from {addr}:
                {msg}")
        s.sendto(data, addr)
    s.close()
    print("[SERVER] Stopped.")
def udp-client():
    c = socket.socket(
        socket.AF_INET,
        socket.SOCK_DGRAM
    )
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to
                stop): ")
```

```
c.sendto(msg.encode(), server)
if msg.lower() == 'quit':
    print("[CLIENT] closing connection")
    break
data, _ = c.recvfrom(1024)
print("Echo from server:", data.decode())
c.close()
print("[CLIENT] stopped")

if __name__ == "__main__":
    threading.Thread(
        target = udp.server,
        = true
    ).start()
    udp_client()
```

OUTPUT:

```
python practical12_udp.py
Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.7:12000
Computer Networks
[SERVER] Received from ('127.0.0.1', 55641): Computer Networks
Echo from server: Computer Networks
Enter message (or 'quit' to 'stop') : quit
[CLIENT] closing connection....
[SERVER] client ('127.0.0.1', 55641) ended the connection
[CLIENT] Stopped
[SERVER] Stopped.
```

RESULT:

Thus end-to-end communication is implemented successfully at transport layer.